

BH Thermodynamics Program Paper 9: Geometric Origin of the 0.036 Drift in the Self-Consistency Equation

— Physical Interpretation via 2D Surface Vibration Modes in a 4D Hyperball

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Nature of this Paper

This is an observational/interpretive paper, not a first-principles proof paper.

We present one interpretation of the physical origin of the coefficient $\pi^2/2$ in the identity $\alpha^{-1} = 137 + (\pi^2/2)\alpha$ of Paper 7 [BH7], based on the uncertainty principle of standard quantum mechanics and dimensional analysis of vibration modes.

This paper does **not** claim:

- That $\pi^2/2$ admits no interpretation other than the one given here
- That the high-precision residual of the observed value $\alpha^{-1} = 137.035999\dots$ is fully derived by the present interpretation
- To exclude other possible interpretations
- To “prove”, “strengthen”, or “extend” the claims of Paper 7

What this paper offers is one way of reading. Paper 7’s claims do not depend on this paper. The numerical fact observed in Paper 7 (that $\alpha^{-1} = 137 + (\pi^2/2)\alpha$ holds at 8.7 ppb precision) is unchanged by whether one adopts this interpretation.

Scope

The scope of this paper:

- **Subject:** $\alpha^{-1}(Q^2 \rightarrow 0) = 137.036$ (geometric origin of in the Thomson limit)
- **Method:** Standard QM uncertainty principle + dimensional analysis of vibration modes in 4D space
- **Conclusion:** The 0.036 drift can be interpreted as the zero-point expectation value of 2D surface vibration modes

This paper does **not** address:

- The high-energy running of ($\alpha^{-1}(M_Z) \approx 127.95$)
- Geometric rederivation of vacuum polarization from Standard Model lepton/quark loops
- Unification of coupling constants (with α_s , α_w , etc.)
- Connection with gravity

These are already well-described by Standard Model vacuum polarization theory, and the present geometric interpretation takes the position of providing a **boundary condition (value at $Q^2 = 0$)** to that framework. The geometric reformulation of the high-energy side is left as future work (§8).

Abstract

In Paper 7 [BH7], the self-consistent equation using unit-cube packing in a 4D ball,

$$\alpha^{-1} = 137 + \frac{\pi^2}{2}\alpha,$$

was reported to agree with the observed value $\alpha^{-1}(0) = 137.035999084$ at a relative precision of 8.7 ppb. This paper presents the following physical interpretation of the structure of this equation:

1. **Although is dimensionless, its physical contributions appear through area-dimensional quantities (§2):** Since the scattering cross section $\sigma_T \propto \alpha^2$, the physical role of is mediated by area.
2. **Classifying 4D space vibration modes by dimension, only 2D face modes can contribute to under isotropic averaging (§3):** 0D/1D/3D/4D modes carry directionality and vanish under averaging, while 2D area remains as a scalar quantity.
3. **The position phase space measure of 2D faces of 137 hypercubes distributed in a 4D unit ball equals $\int_{B_4(1)} dV = \pi^2/2$ (§4):** While this coincides numerically with $V_4(1)$, in our interpretation it is read as “the integral of position degrees of freedom of 2D faces”.
4. **The W7 self-consistent equation $\alpha^{-1} = 137 + (\pi^2/2)\alpha$ can be interpreted as a self-consistent correction from zero-point vibrations of 2D face modes of the 137 hypercubes (§5).**
5. **This structure is identical to the plaquette action of Wilson lattice gauge theory (§6):** The isomorphism shown in Paper 8 [BH8] is realized at the geometric level.
6. **There exists a theoretical lower bound on the observed value of , arising from the width of the 2D mode distribution (§7).**

Our interpretation is completed within the framework of standard quantum mechanics and does not require new assumptions (discrete spacetime, complex integer lattices, etc.).

§1 Introduction

Paper 7 [BH7] presented the self-consistent equation

$$\alpha^{-1} = N(1) + V_4(1)\alpha = 137 + \frac{\pi^2}{2}\alpha$$

using the 137 unit 4D cubes (tesseract) inscribed in the 4D ball $B_4(R=3)$ and the volume of the 4D unit ball $V_4(1) = \pi^2/2$. This equation agrees with the observed value $\alpha^{-1}(0) = 137.035999084(21)$ at a precision of 8.7 ppb.

The structure of this self-consistent equation left the following unresolved questions:

Question 1: Why does $\pi^2/2$ appear as a coefficient of ? - Numerically it coincides with $V_4(1)$, but it is physically unclear why the volume of a 4D unit ball should appear - Is it a mere mathematical identity, or does it carry physical meaning?

Question 2: Geometric origin of the 0.036 drift - The geometric structure giving the integer 137 (W6 paper) is clear - But the origin of “0.036” was only explained as a closure of the self-consistency

Question 3: Physical content of the isomorphism with Wilson lattice gauge theory shown in W8 [BH8] - Structural correspondence is established - But the physical content of the correspondence (why it holds) remains unclear

This paper presents the following interpretation in response to these questions:

is physically an area-dimensional quantity ($\sigma \propto \alpha^2$), and only 2D face modes can contribute to among the 4D space vibration modes. The position phase space measure of 2D face vibration modes of the 137 hypercubes gives $\pi^2/2$, which physically explains the coefficient structure of the W7 self-consistent equation.

This interpretation takes the standard QM uncertainty principle as its starting point and requires no new assumptions.

§2 Dimensional Analysis of and Area Dimensionality

2.1 itself is dimensionless

The fine-structure constant is defined in SI units as

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} \approx 7.297 \times 10^{-3}$$

and is dimensionless.

2.2 Physical contributions appear through α^2

However, in many cases the quantities through which is physically involved appear as α^2 :

Thomson scattering cross section:

$$\sigma_T = \frac{8\pi}{3} r_e^2 = \frac{8\pi}{3} \left(\alpha \cdot \frac{\hbar}{mc} \right)^2 \propto \alpha^2 \cdot [\text{length}]^2$$

Rutherford scattering differential cross section:

$$\frac{d\sigma}{d\Omega} \propto \alpha^2 \cdot [\text{length}]^2$$

Bohr radius:

$$a_0 = \frac{\hbar}{\alpha mc} \propto \alpha^{-1} \cdot [\text{length}]$$

Classical electron radius:

$$r_e = \alpha \cdot \frac{\hbar}{mc} \propto \alpha \cdot [\text{length}]$$

Thus, is a quantity directly linked to **cross section (area dimension)**.

2.3 Correspondence with Geometric Structures in 4D Space

When considering geometric structures in 4D space to which naturally couples:

- (cross section) $\propto$ area
- $\rightarrow$ is a quantity that couples to **2-dimensional structures**

This is consistent with the fact that in Wilson lattice gauge theory, the gauge coupling appears as the action coefficient of plaquettes (2D faces, 2-forms).

2.4 Starting Point of This Paper

From the above:

When considering contributions to from vibration modes in 4D space, the area dimensionality of restricts the contributing modes to those with “2-dimensional area”.

This is the central observation of this paper.

§3 Dimensional Classification of Vibration Modes in 4D Space

3.1 Vibration Modes of 137 Hypercubes and the Circumscribed 4-Sphere

The system treated in Paper 7: - 4D ball $B_4(3)$ of radius $R = 3$ - 137 unit 4D cubes (tesseract) packed inside

This system has the following vibration modes:

Internal structure of one tesseract:

Structural element	Count	Independent vibration modes
Vertex (0-cell)	16	$16 \times 4 = 64$ directions
Edge (1-cell)	32	Stretching 32, transverse, etc.
Face (2-cell, square)	24	Bulging, twisting
Solid (3-cell, cube)	8	Volume vibration
4-cell (whole tesseract)	1	Whole breathing mode

Across all 137 tesseracts: - Each structural element’s vibration multiplied by 137 - Interaction modes between adjacent cubes - Phonon-like spectrum as a whole

Circumscribed 4-sphere S^3 (radius 3):

Mode	Description
Radial breathing	Vibration of $R = 3$
Spherical harmonics $Y_{\ell mn}$ on S^3	Angular vibrations ($\ell = 0, 1, 2, \dots$)
4-axis distortion	Sphere $\rightarrow$ ellipsoid deformation

3.2 Survival Conditions under Isotropic Averaging

Among these modes, what matters for contributions to is the **survival condition under isotropic averaging**.

is dimensionless and scalar, invariant under $SO(4)$ symmetry. Therefore contributions to are limited to **modes that do not vanish under isotropic averaging (averaging over 4-axis directions)**.

Mode dimension	Mathematical property	Result of isotropic averaging
0D (point)	Vector position fluctuation	Zero (translation symmetry)
1D (line segment)	Has direction	Zero (line symmetry, sign flips on reversal)
2D (face)	Area (scalar quantity, no direction)	Survives
3D (volume, oriented)	Volume element $dV = dx \wedge dy \wedge dz$ is pseudoscalar (has orientation)	Zero (sign flips on reversal)
4D (4-volume)	4-volume element is pseudoscalar	Zero

Key fact: Area is a scalar quantity, and area is invariant under flipping the orientation of the face. In contrast, length and volume (oriented volume elements) change sign under reversal of orientation.

3.3 Selection Principle

From the above:

Among vibration modes in 4D space, only 2D face modes can give significant contributions to under isotropic averaging.

This is consistent with the area dimensionality of (§2).

3.4 Consistency with Wilson Lattice Gauge Theory

This conclusion agrees with the structure of **2-form gauge fields (plaquette action)** in Wilson lattice gauge theory:

- 0-form (scalar field): gauge-invariant point-like structure
- 1-form (vector potential A_μ): gauge-dependent
- **2-form (field strength $F_{\mu\nu}$): gauge-invariant, plaquette action $S = \beta \sum_p \text{Re tr}(U_p)$**
- 3-form, 4-form: higher-order structures

That “couples to faces” is a standard fact in Wilson formalism.

§4 Position Phase Space of 2D Face Modes of 137 Hypercubes

4.1 Position Degrees of Freedom of 2D Face Modes

To fully specify a 2D face vibration mode requires:

- **Position:** Where the 2D face is located in the 4D ball $B_4(R=3)$
- **Direction:** One of 6 possible 2-plane directions in 4D space ($\binom{4}{2} = 6$)
- **Amplitude:** Excitation level of the vibration

4.2 Integration of Position Degrees of Freedom

For a fixed direction and amplitude, integrating the position degrees of freedom of 2D faces within the 4D unit ball (scale-normalized) gives:

$$\text{2D face position phase space} = \int_{B_4(1)} dV = \frac{\pi^2}{2}$$

This coincides numerically with $V_4(1) = \pi^2/2$.

4.3 Interpretation: Reading of Physical Meaning

Interpretation	Physical meaning of $\pi^2/2$
Old interpretation (W7)	Volume $V_4(1)$ of 4D unit ball
This paper	Measure of position degrees of freedom for placing 2D face modes in 4D unit ball

The two are **numerically identical, but physically different**. Under our interpretation, $\pi^2/2$ has a necessary reason to appear as the coefficient of : since couples to 2D face modes, the volume of their position phase space appears as the geometric coefficient of the coupling constant.

4.4 Direction and Amplitude Contributions

The direction contribution from 6 possible 2-planes may appear as a combinatorial factor separately. In this paper, we treat the position integral as the dominant factor, and consider the directional factor as an $O(1)$ correction even if numerically effective.

The amplitude appears as itself (excitation probability amplitude):

$$\text{Collective contribution of 2D face modes} = \underbrace{\frac{\pi^2}{2}}_{\text{position phase space}} \times \underbrace{\alpha}_{\text{amplitude per face}} = \frac{\pi^2}{2} \alpha$$

This physically explains the coefficient term in the W7 self-consistent equation.

§5 Physical Interpretation of the W7 Self-Consistent Equation

5.1 Structure of the Equation

The self-consistent equation of Paper 7 [BH7]:

$$\alpha^{-1} = 137 + \frac{\pi^2}{2} \alpha$$

5.2 Each Term under Our Interpretation

Term	Mathematical content	Physical interpretation (this paper)
137	Number of unit cubes packed in 4D ball $B_4(3)$	Contribution when all vibration modes are in the ground state (lowest energy): integer-theoretic fact of 4D lattice
$\pi^2/2$	$V_4(1)$, or in our interpretation, 2D face mode position phase space	Position degrees of freedom for placing 2D face modes of 137 hypercubes in the 4D unit ball
(in coefficient term)	Coupling constant	Excitation amplitude of each 2D face mode
$(\pi^2/2)\alpha$	0.036 (numerically)	Collective contribution of zero-point vibrations of 2D face modes (from standard QM uncertainty principle)

5.3 Connection with Standard QM Uncertainty Principle

Each 2D face mode has the following zero-point energy under the harmonic oscillator approximation:

$$E_{\text{zero-point}}^{(i)} = \frac{1}{2} \hbar \omega_i$$

where ω_i is the eigen-frequency of the i-th 2D face mode.

The collective contribution of all 2D face modes is:

$$\sum_i \frac{1}{2} \hbar \omega_i = (\text{position phase space measure}) \times (\text{amplitude density}) = \frac{\pi^2}{2} \cdot \alpha$$

Here the α on the right side represents the self-consistency condition that the average excitation amplitude of each mode equals itself.

5.4 Self-Consistency

From the structure that α is the amplitude of 2D face modes and that the zero-point contribution of 2D face modes provides a correction to α^{-1} , the self-consistent equation

$$\alpha^{-1} = 137 + \frac{\pi^2}{2} \alpha \quad \Leftrightarrow \quad \frac{\pi^2}{2} \alpha^2 + 137\alpha - 1 = 0$$

is naturally derived.

5.5 Numerical Agreement

The positive root of this equation:

$$\alpha^{-1} = 137.036010988 \dots$$

The relative error with the CODATA 2018 value $\alpha^{-1}(0) = 137.035999084(21)$ is **8.7 ppb**. This is the fact already observed in Paper 7, and our interpretation provides a **physical framework** explaining this numerical agreement.

§6 Strengthening of the Isomorphism with Wilson Lattice Gauge Theory

Paper 8 [BH8] showed that there is a structural correspondence between the W7 self-consistent equation and Wilson lattice gauge theory based on Schläfli duality and B_4 equivariance. Our 2D face mode interpretation clarifies the **physical content** of this correspondence.

6.1 Correspondence

Wilson lattice gauge theory	W7 self-consistency (this paper's interpretation)
Coupling constant $\beta = 1/g^2$	α^{-1}
Plaquette (2-form structure)	2D face vibration mode
Number of plaquettes $\sum_p$	$\pi^2/2$ (position phase space measure of 2D faces)
Plaquette action $\beta \sum_p \text{Re tr}(U_p)$	$(\pi^2/2)\alpha$
Cell (4D hard core)	137 (number of unit cube packing)
Total action	$\alpha^{-1} = 137 + (\pi^2/2)\alpha$

6.2 Clarification of Physical Content

The isomorphism in Paper 8 was established as a **structural correspondence**, but with this paper it is **physically positioned** as:

The W7 self-consistent equation is the realization of the plaquette action structure of Wilson lattice gauge theory on the 4D ball geometry.

6.3 Implications

With this: - The correspondence between W7 and W8 is elevated from “mathematical isomorphism” to “physical isomorphism” - coupling to “2D faces” aligns with the natural interpretation of Wilson formalism - A roadmap appears for **geometric reconstruction** of vacuum polarization in standard QFT

§7 Theoretical Lower Bound on Observed Width of

7.1 Distribution Width of Zero-Point Vibrations

Each 2D face mode's zero-point vibration has a finite distribution width from the standard QM uncertainty principle:

$$\Delta n_i^{\text{zero}} = \frac{1}{2}$$

(zero-point fluctuation of the photon number operator)

7.2 Propagation to

The fluctuation in the collective contribution of all 2D face modes is:

$$(\Delta\alpha^{-1})^2 \approx \sum_i \left(\frac{\partial\alpha^{-1}}{\partial n_i} \right)^2 (\Delta n_i)^2$$

A concrete calculation is beyond the scope of this paper (requires determination of mode eigen-frequencies), but structurally:

There exists a theoretical lower bound on the observed value of , arising from the width of the 2D face mode distribution.

7.3 Correspondence with CODATA Measurement

$$\alpha^{-1}(0) = 137.035999084(21)$$

The CODATA uncertainty (21) $\approx 2.1 \times 10^{-8}$ is experimental precision, and whether our theoretical lower bound is even smaller requires further consideration of mode frequencies.

7.4 Implications

By our interpretation: - is not a “sharp point” but the “center value of a distribution” - The observed 8.7 ppb deviation (W7) leaves room for higher-order mode contributions - Future precision measurements may make our theoretical lower bound observable

§8 Open Problems: Extension to High Energies

Our interpretation **addresses only the Thomson limit** $\alpha^{-1}(Q^2 \rightarrow 0) = 137.036$.

8.1 Running of in Standard QED

In the Standard Model, runs with the energy scale Q^2 :

- $\alpha^{-1}(0) = 137.036$ (Thomson limit)
- $\alpha^{-1}(M_Z^2) \approx 127.95$ (Z boson mass scale)
- The difference ≈ 9.08 units is explained by vacuum polarization from lepton/quark loops

This running is **well-described** by precision measurements of standard QED, and this paper does not compete with it.

8.2 Position of This Paper

Our geometric interpretation provides a physical origin for the **boundary condition** (value at $Q^2 = 0$) of . Running is left to standard QED.

The two are hierarchically complementary:

Layer	Content	Responsibility
1	137 (integer, geometric invariant)	Paper 6 [BH6]: integer theory of packing
2	$(\pi^2/2)\alpha \approx 0.036$	This paper: 2D face modes
3	$\Delta\alpha_{SM}(Q^2)$	Standard QED vacuum polarization

8.3 Future Tasks

The relationship between layer 3 (standard QED vacuum polarization) and our geometric interpretation leaves the following open questions:

1. **Additional modes activated at high energies:** 3D and 4D vibration modes may start contributing to at high energies
2. **Geometric counterparts of SM loops:** Whether each of electron loop, muon loop, quark loop corresponds to a specific geometric mode
3. **Geometric derivation of $\alpha^{-1}(M_Z) \approx 128$:** Whether our framework can be extended to high energies
4. **behavior at GUT scale:** Geometric prediction of coupling constants at grand unification scale

The geometric reformulation of these is left as future research. **The claim of this paper (geometric origin of the Thomson limit) holds independently of the mechanism of running.**

§9 Conclusion

This paper presented the following as a physical interpretation of the self-consistent equation $\alpha^{-1} = 137 + (\pi^2/2)\alpha$ in Paper 7 [BH7]:

1. **is dimensionless but physically area-dimensional** ($\sigma \propto \alpha^2$)
2. **Among 4D space vibration modes, only 2D face modes can contribute to under isotropic averaging**
3. **Position phase space measure of 2D faces of 137 hypercubes distributed in the 4D unit ball = $\pi^2/2$**
4. **The W7 self-consistent equation can be interpreted as a self-consistent contribution of zero-point vibrations of 2D face modes**
5. **This structure is identical to Wilson lattice gauge theory plaquette action** (physical content of W8 isomorphism)
6. **The observed value of has a theoretical lower bound arising from the 2D mode distribution width**

Our interpretation is completed within the framework of the standard QM uncertainty principle, and requires no new assumptions (discrete spacetime, extra dimensions, etc.).

The scope of this paper is only the Thomson limit of . The high-energy running is left to standard QED, and its geometric reformulation is left as future tasks (§8).

Position of This Paper

Relationship among Paper 7 [BH7], Paper 8 [BH8], and this paper:

Paper	Content	Area
BH7 (identity)	Discovery of $\alpha^{-1} = 137 + (\pi^2/2)\alpha$	Algebraic observation
BH8 (Wilson isomorphism)	Proof of structural correspondence	Mathematical correspondence
This paper (BH9)	Physical interpretation via 2D face modes	Physical content

With this trilogy, the geometric origin of is positioned in three layers: **observation** → **structure** → **physical content**.

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Revision History

- **v1 (May 20, 2026)**: Initial version